Examiner's Use

7 An induced nuclear fission reaction may be represented by the equation

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n$$

(a) Sketch the variation with nucleon number of the binding energy per nucleon in Fig 7.1. [2]

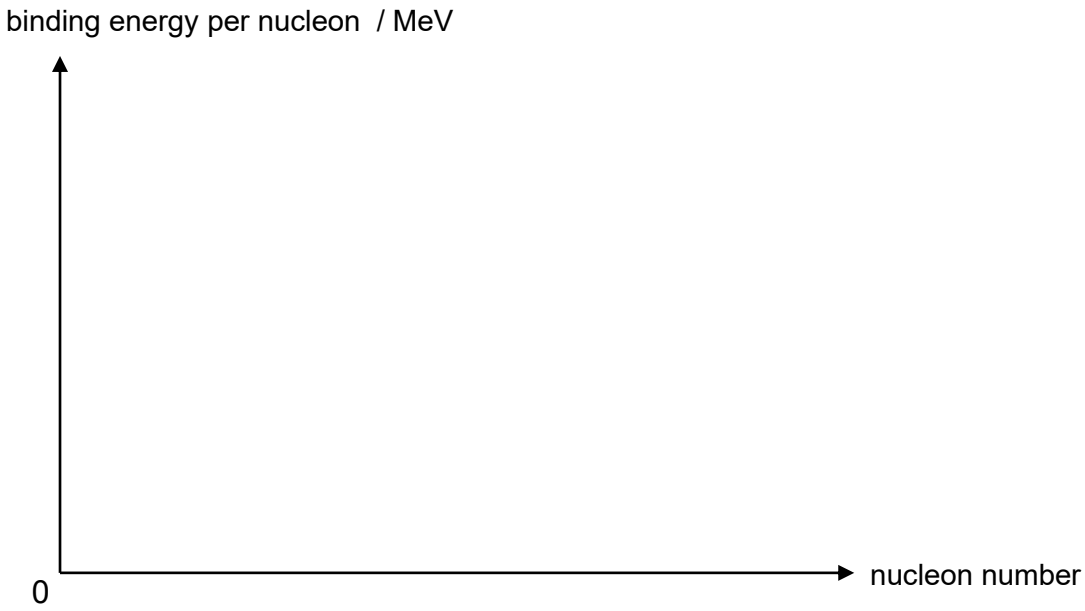

Fig 7.1

(b) Hence, explain why Uranium – 235 is more likely to undergo fission than fusion.

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……[3]

For Examiner's Use

(c) The masses of the various nuclides are as listed below:

nuclide	mass / *u*
Uranium - 235	235.04393
Barium – 141	140.91441
Krypton - 92	91.92616
neutron	1.00867

Determine the energy released in the reaction.

energy released = ... J [2]